# Bartosz Osiej

**Rust Systems Engineer — consensus, schedulers & kernel-edge performance**

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Szczecin, Poland · Remote (CET) · available immediately

## SUMMARY

Rust engineer who builds low-level infrastructure from scratch and lives in the hot path. Author of **TrustNode** — a custom Solana-style validator implementing PoH, Tower BFT, a Sealevel-style parallel scheduler, Turbine propagation and CRDS gossip as a 10-crate workspace. Also ship kernel-level eBPF security tooling and compilers — with strict lint gates, cargo audit and CI as defaults.

## FLAGSHIP PROJECT

### TrustNode — Solana-style validator in Rust **from scratch**

github.com/BartoszOsiej/TrustNode · public · Apache-2.0

- **Sealevel-style scheduler** — conflict analysis over read/write account sets, non-conflicting transactions grouped into parallel batches, write-locks serialized; slot-based batches with compute budget.
- **Tower BFT consensus** — PoH-verified fork choice with lockout thresholds and 2/3 supermajority finality.
- **PoH history** — hash-chained verifier stream with tick scheduling and entry verification.
- **Turbine propagation** — tree-based block shred distribution with Reed-Solomon erasure coding (encode/decode verified against k-of-n recovery in tests).
- **CRDS gossip** — push/pull membership protocol, contact-info windows, duplicate/shred detection.
- **Persistent blockstore** — append-only, hash-chained frames, torn-write recovery on replay; JSON-RPC server (slot, accounts, balances, cluster nodes).
- **Engineering gates:** 127 tests across the workspace · `cargo clippy -D warnings` · rustfmt · `cargo audit` (zero advisories) · rustdoc `-D warnings` · GitHub Actions CI, CodeQL and OSSF Scorecard.
- *Microbenchmarks built in (`--bench`): PoH verification throughput, account store insert/lookup, erasure-coding encode/decode, transaction execution.*

## OTHER SELECTED WORK **(all public on GitHub)**

### Talus Process Monitor — anti-ransomware in the Linux kernel

Rust + eBPF · crates.io + PyPI

*In-kernel syscall interception (execve, openat, network egress) with an in-kernel sliding-window verdict that SIGKILLs mass-encryption processes — no user-space round-trips. ~500k events/s through per-CPU perf buffers; automatic response <1s; shipped packages, 4 CI pipelines.*

### Scryer — lead-intelligence pipeline

Rust · deterministic ETL

*Load → validate → dedupe → score → enrich → rank → export; DNS parser tested on frozen real-world fixtures; outputs signed with SHA-256 manifests — reproducible, audit-ready (62 tests).*

### Externum

own language

*Compiler from scratch targeting readable Python, Bash and VM bytecode; differential harness compares all backends in CI (~380 tests).*

### Post-quantum crypto + LSM

Rust / C

*ML-KEM-768 + AES-256-GCM file-encryption CLI (fuzzed) and a stackable Linux Security Module with upstream-style patches, runnable in QEMU.*

## SKILLS

| | |
|---|---|
| **Core** | Rust (workspace design, async/tokio, unsafe auditing), C, Python, Bash |
| **Systems** | Linux (daily driver), eBPF/aya-rs, perf buffers, networking (UDP/TCP, RPC), scheduling & concurrency, serde/JSON, hash chains & erasure coding |
| **Blockchain** | Solana architecture: PoH, Tower BFT, Sealevel, Turbine, CRDS gossip, block builders & transaction landing (theory + working implementation) |
| **Quality** | clippy -D warnings, rustfmt, cargo audit / cargo-deny, CodeQL, GitHub Actions CI, differential & fuzz testing, SLSA provenance / cosign (Talus) |
| **Also** | Docker, GitHub Actions, Nginx, post-quantum crypto (ML-KEM), full-stack TS/Python when a tool needs a UI |

## EXPERIENCE

### Founder — Hartwell Labs · independent software studio · 2025–present

- Own products end-to-end (design → tests → CI → releases → docs); commercial side too: contracts, pricing, invoicing. Client work: security assessments (under contract), integrations, Linux hardening — remote, async, English.

### Independent engineer & open-source author · 2022–present

- 40+ public repositories (Rust, C, Python, TypeScript), every one with tests and CI; packages published to crates.io and PyPI.

## EDUCATION

Independent study since 2022 — systems, security, compilers, consensus — proven in shipped projects and public repos.

## LANGUAGES

**Polish** — native · **English** — B2+ (technical documentation, written correspondence, daily async work)